Extraction and Organisation of Statistical Vocabularies from Eurostat Tables

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Abstract

Statistical tables contain a large amount of domain-specific vocabulary distributed across titles, headers and descriptive columns. However, this vocabulary is not normally provided as a structured resource and cannot be directly reused for semantic search, metadata management or statistical data integration. This project presents an automatic pipeline for extracting and organising statistical vocabulary from a large collection of Eurostat tables.

For each table, the system identifies temporal intervals, extracts non-numeric descriptive strings, detects geographical terms using the NUTS nomenclature and processes the associated title. The resulting terms are combined into a global vocabulary and classified into measures, dimension names, dimension values, units and other terms. The extracted measures are subsequently assigned to seventeen statistical domains using a hybrid approach based on lexical rules and TF-IDF similarity.

The final execution processed [**NUMBER OF SUCCESSFULLY PROCESSED TABLES**] tables and produced a global vocabulary containing [**NUMBER OF TERMS IN THE FINAL VOCABULARY**] distinct terms. The vocabulary classification obtained an accuracy of [**CLASSIFICATION ACCURACY**], while the domain assignment achieved an accuracy of [**DOMAIN ASSIGNMENT ACCURACY**] on manually annotated samples. The results show that the proposed pipeline can transform heterogeneous statistical tables into a structured and reusable vocabulary while remaining computationally feasible for a large dataset.

1 Introduction

Statistical institutions publish large collections of tables covering areas such as population, employment, education, health, transport, agriculture, industry and the environment. These tables contain valuable terminology, but the relevant concepts are distributed across different structural elements. Measures can appear in titles or headers, dimensions are usually represented by descriptive columns, units may be embedded in labels, and geographical entities are often mixed with other categorical values.

This organisation is suitable for human interpretation but makes automatic reuse more difficult. A person can easily recognise that a table contains an unemployment rate organised by age, sex and country. A software system, however, must first distinguish the measure from its dimensions, dimension values, units, temporal references and geographical entities. Without this separation, it is difficult to compare concepts between tables, construct reusable vocabularies or provide semantic access to statistical datasets.

The objective of this project is to extract the textual content of Eurostat tables and transform it into a structured statistical vocabulary. Instead of treating every table as an isolated file, the proposed pipeline identifies its semantic components, normalises them and combines the results across the complete collection.

For every table t, the system first extracts the temporal expressions that form $D(t)$. It then identifies the leftmost descriptive columns and obtains the duplicate-free set $S(t)$ of non-numeric strings. Geographical terms are detected using a dictionary constructed from the NUTS nomenclature and removed from the provisional vocabulary:

$$V(t) = S(t) \setminus Geo(t) \tag{1}$$

The corresponding table title is also processed. Dates and geographical references are removed, while the remaining semantic content is added to $V(t)$. The individual table vocabularies are then combined into the global vocabulary:

$$V = \bigcup_t V(t) \tag{2}$$

The global vocabulary is classified into five categories: measures, dimension names, dimension values, units and other terms. The extracted measures are subsequently grouped into seventeen statistical domains. Since both stages require semantic interpretation, their quality is evaluated using manually annotated samples.

The project also addresses the technical difficulties of processing a large and heterogeneous corpus. The complete dataset contains thousands of tables, several internally compressed files and individual tables large enough to make direct in-memory processing impractical. For this reason, the implementation uses flexible CSV detection, automatic GZIP decompression and block-based reading of the descriptive columns.

The main contributions of the project are therefore:

- an end-to-end pipeline for extracting and organising vocabulary from statistical tables;
- a scalable procedure for processing large and internally compressed files;
- a structured classification into measures, dimensions, values and units;
- a hybrid method for assigning measures to statistical domains;
- a manual evaluation of the two main semantic classification stages.

The remainder of the report follows the sequence of the implemented pipeline. It first presents the input resources and the project architecture. The following sections describe temporal extraction, vocabulary extraction, geographical recognition, title processing and construction of the global vocabulary. The report then explains vocabulary classification, measure-domain assignment, evaluation, implementation decisions, limitations and final conclusions.

2 Dataset and Project Architecture

2.1 Input resources

The project uses three main input resources: the statistical tables, the mapping between filenames and table titles, and the NUTS geographical nomenclature.

The main dataset contains 7,605 Eurostat statistical tables stored as CSV files. Although the files use the `.csv` extension, some of them are internally compressed with GZIP. The collection contains tables of very different sizes, from small files with a limited number of rows to individual tables close to one gigabyte.

Two versions of the table collection were used during development. A reduced dataset containing 2,000 tables was used to implement and test the first versions of the pipeline. This made it possible to identify errors and adjust the extraction and classification rules without repeatedly processing the complete collection. Once the pipeline was stable, the final execution was performed on the full dataset.

The second resource is a CSV file that associates each table filename with its corresponding Eurostat title. These titles provide additional semantic information that may not appear explicitly in the descriptive columns of the table. They are therefore processed and incorporated into the vocabulary after removing temporal and geographical references.

The third resource is the NUTS 2021–2024 workbook. This file contains European geographical codes and names at different territorial levels. It is used to build a geographical dictionary that supports the identification and removal of geographical entities from the statistical vocabulary.

Table 1 summarises the main resources used by the pipeline.

Table 1: Input resources used by the project.

Resource	**Content**	**Purpose**
Eurostat tables	Statistical data, headers and descriptive columns	Extraction of dates, terms, measures, dimensions and units
Title mapping	Association between filenames and official table titles	Extraction of additional semantic vocabulary
NUTS workbook	European geographical codes, names and territorial levels	Identification of geographical terms

2.2 General processing workflow

The system is implemented as a sequential pipeline. Each stage receives the results produced by the previous stage and generates intermediate CSV files that can be inspected independently.

The general workflow is:

`Statistical tables` → `Temporal and textual extraction` → `Geographical filtering` → `Title processing` → `Global vocabulary` → `Vocabulary classification` → `Domain assignment`

The first stage examines the header of every table and extracts temporal expressions. The second stage identifies the descriptive columns and extracts their non-numeric string values. Geographical terms are then detected using the NUTS dictionary and removed from the provisional vocabulary.

After this filtering, the official title associated with each table is processed. Dates and geographical references are removed, and the remaining title content is added to the table vocabulary. All table vocabularies are subsequently merged into a single global vocabulary.

The final stages classify every term into a semantic category and assign each detected measure to a statistical domain. Intermediate and final results are stored separately so that every transformation can be analysed and validated.

2.3 Software organisation

The implementation is divided into independent Python modules. Each module is responsible for one part of the pipeline:

- `extract_time.py` extracts temporal intervals.
- `extract_vocabulary.py` identifies descriptive columns and extracts textual terms.
- `extract_geo.py` builds the geographical dictionary and detects geographical entities.
- `process_titles.py` processes the official table titles.
- `classify_terms.py` assigns terms to the vocabulary categories.
- `cluster_measures.py` assigns measures to statistical domains.
- `evaluate.py` computes the evaluation metrics.
- `pipeline.py` coordinates the complete execution.

The main entry point is `main.py`. The behaviour of the pipeline is controlled through a YAML configuration file. This file defines the dataset mode, input and output directories, chunk size, random seed and optional exclusions.

All paths are relative to the project directory. This avoids dependencies on the original development computer and makes the repository reproducible on a different system.

2.4 Output organisation

The generated files are divided into intermediate and final outputs. Intermediate files preserve the results of each extraction step, while final outputs contain the classified vocabulary and the domain assignments.

The most relevant intermediate files include:

- temporal intervals extracted from each table;
- table-level textual vocabulary;
- detected geographical terms;
- processed titles;
- final global vocabulary.

The final output directory contains:

- the complete classification of the vocabulary;
- separate files for measures, dimension names, dimension values, units and other terms;
- the domain assigned to every measure;
- summaries and representative examples for each domain.

This separation improves traceability because the result of every stage can be checked without rerunning the entire pipeline. It also simplifies error analysis and allows the final vocabulary to be reused independently of the original tables.

3 Extraction of Temporal Intervals

3.1 Objective

The first stage identifies the temporal information associated with each statistical table. This information forms the set $D(t)$ for a table t.

Temporal expressions are mainly located in the table header and may appear as individual years, ranges or labels that combine several periods. Detecting them separately prevents dates from being included later as statistical vocabulary.

3.2 Extraction method

For each table, the system reads only the header instead of loading the complete file. This is sufficient because the temporal dimension is represented in the column names.

The extraction procedure performs the following operations:

1. Read the table header using flexible delimiter, encoding and compression detection.
2. Search the column labels for individual years and temporal ranges.
3. Validate the detected years using the limits defined in the configuration.
4. Normalise the expressions and remove duplicates.
5. Store the resulting intervals together with the corresponding table name.

The accepted year range was configured between 1900 and 2100. This reduces false detections caused by numerical codes or identifiers that are not temporal values.

The output preserves both the original temporal expression and its normalised interpretation. A summary file also records whether every table was processed correctly and how many intervals were detected.

3.3 Results

The final execution selected [**NUMBER OF TABLES SELECTED**] tables for processing. Temporal extraction completed successfully for [**NUMBER OF TABLES PROCESSED IN STEP 1**] tables, while [**NUMBER OF TABLES WITH ERRORS IN STEP 1**] tables produced errors.

A total of [**NUMBER OF TEMPORAL INTERVALS**] temporal expressions were extracted. In addition, [**NUMBER OF TABLES WITHOUT DATES**] tables did not contain a detectable year or temporal interval in their header.

Table 2 summarises the final results.

Table 2: Results of temporal interval extraction.

Result	**Value**
Selected tables	7,604
Successfully processed tables	7,604
Tables with errors	0
Tables without detected dates	31
Extracted temporal intervals	198,090

This stage is computationally inexpensive because it only reads the first row of every table. Its main purpose is to separate temporal metadata from the vocabulary extracted in the following stages.

4 Extraction of the Table Vocabulary $S(t)$

4.1 Objective

The second stage extracts the descriptive strings contained in every table. For a table t, the resulting set $S(t)$ contains the distinct non-numeric strings found in the descriptive columns.

Statistical tables usually combine descriptive columns on the left with numerical observation columns on the right. The descriptive columns contain categories such as sex, age group, activity, product, occupation or unit. These values represent the main source of vocabulary.

The objective is therefore to identify the descriptive part of each table and extract its textual content without processing the numerical observations as vocabulary.

4.2 Identification of descriptive columns

The pipeline analyses the columns from left to right. A column is considered descriptive when its header and values mainly contain textual information rather than numerical observations.

The descriptive region normally ends when the temporal or numerical observation columns begin. Once this boundary is identified, only the columns located before it are used for vocabulary extraction.

The number of descriptive columns may vary considerably between tables. Some tables contain a single category column, while others combine several dimensions such as geography, sex, age, activity and unit.

Column headers are also considered useful vocabulary because they often represent dimension names. Consequently, both the names of the descriptive columns and their textual values are included in the extraction process.

4.3 String normalisation

Before a string is included in $S(t)$, it is normalised to reduce superficial differences between equivalent expressions.

The normalisation process includes:

- removal of unnecessary leading and trailing spaces;
- replacement of repeated internal whitespace;
- rejection of empty values;
- rejection of purely numeric values;
- rejection of detected temporal expressions;
- preservation of meaningful punctuation and abbreviations;
- elimination of duplicates within the same table.

Each extracted term is stored together with its source table, frequency and type of origin, such as column header or column value. This information is later used as contextual evidence during classification.

The global provisional vocabulary is constructed by combining the distinct terms obtained from all tables.

4.4 Block-based processing

Some tables are too large to be loaded completely into memory. For this reason, vocabulary extraction is performed incrementally.

After identifying the descriptive columns, the table is read in blocks of [**CHUNK SIZE**] rows. Only the descriptive columns are loaded, while the numerical observation columns are ignored.

For every block, the system:

1. extracts the values of the selected columns;
2. normalises the strings;
3. updates their occurrence counts;
4. removes duplicates through table-level dictionaries;
5. releases the block before reading the next one.

This approach limits memory consumption and makes it possible to process large files using a conventional computer. It is especially relevant because several files are internally compressed and must be decompressed while they are being read.

4.5 Results

The final execution processed [**NUMBER OF SUCCESSFULLY PROCESSED TABLES IN STEP 2**] tables successfully during vocabulary extraction. A total of [**NUMBER OF TABLES WITH ERRORS IN STEP 2**] tables could not be processed.

The extraction generated [**NUMBER OF DISTINCT TABLE-TERM PAIRS**] distinct table-term pairs. After combining repeated terms across tables, the provisional global vocabulary contained [**NUMBER OF DISTINCT GLOBAL TERMS**] distinct strings.

Table 3 presents the main results.

Table 3: Results of table vocabulary extraction.

Result	**Value**
Successfully processed tables	7,604
Tables with errors	0
Distinct table-term pairs	1,151,463
Distinct terms in the provisional vocabulary	159,477
Processing block size	20,000 rows

This was the most computationally expensive stage of the pipeline because it required reading the contents of all selected tables. However, the use of block-based processing kept memory usage stable and avoided loading complete tables.

The output of this stage still contains geographical terms. These terms are identified and separated in the following stage using the NUTS nomenclature.

5 Identification of Geographical Terms

5.1 Objective

The vocabulary extracted from the descriptive columns contains many geographical names and codes. These terms are useful as metadata, but they should not be mixed with measures, dimensions or units.

For each table t, the objective of this stage is to identify the geographical subset $Geo(t)$ and remove it from the provisional vocabulary:

$$V(t) = S(t) \setminus Geo(t) \tag{3}$$

5.2 Construction of the geographical dictionary

The geographical dictionary was created from the NUTS 2021–2024 workbook. The source includes territorial codes and official names for different European geographical levels.

The dictionary contains normalised geographical expressions and their corresponding metadata. Several variants may refer to the same geographical entity, including:

- official territorial names;
- NUTS codes;
- upper- and lower-case variants;
- names with normalised punctuation and whitespace;
- alternative forms included in the source workbook.

All dictionary entries are normalised using the same procedure applied to the extracted table terms. This allows geographical matching to be performed independently of minor formatting differences.

5.3 Geographical matching

Each term in $S(t)$ is compared with the geographical dictionary. A term is considered geographical when its normalised form matches a known geographical name or code.

The detected geographical terms are stored separately, while the remaining terms form the provisional non-geographical vocabulary of the table.

The matching process is performed at table level so that the origin of each geographical term is preserved. This makes it possible to analyse which tables contain each location and how often geographical values occur across the collection.

5.4 Results

The geographical dictionary contained [**NUMBER OF GEOGRAPHICAL DICTIONARY ENTRIES**] normalised entries.

The matching process detected [**NUMBER OF GEOGRAPHICAL TABLE-TERM PAIRS**] geographical table-term pairs and [**NUMBER OF DISTINCT GEOGRAPHICAL TERMS**] distinct geographical terms.

After their removal, the remaining vocabulary contained [**NUMBER OF REMAINING TABLE-TERM PAIRS**] table-term pairs and [**NUMBER OF REMAINING DISTINCT TERMS**] distinct global terms.

Table 4 summarises these results.

Table 4: Results of geographical term identification.

Result	**Value**
Geographical dictionary entries	4,512
Geographical table-term pairs	413,625
Distinct geographical terms	3,558
Remaining table-term pairs	737,838
Remaining distinct terms	155,919

The removal of geographical terms reduces the number of dimension values that would otherwise dominate the final vocabulary. It also prevents country and region names from being incorrectly classified as statistical concepts.

6 Processing of Table Titles

6.1 Objective

Table titles contain relevant semantic information that may not appear explicitly in the descriptive columns. In many cases, the title identifies the main measure, the subject of the table or the population being analysed.

The objective of this stage is to associate each table with its official title, remove information already treated separately and add the remaining semantic content to the table vocabulary.

6.2 Title association

The title mapping file links each table filename to its official Eurostat title. The association is performed through the normalised filename.

For every selected table, the system searches for the corresponding title. The execution records whether the title was found and preserves the original text for inspection.

Tables without an associated title can still be processed using the vocabulary extracted from their descriptive columns, although they lose an additional source of semantic information.

6.3 Removal of dates and geographical references

The title is processed using the same temporal and geographical resources applied in the previous stages.

First, years and temporal expressions are detected and removed. Geographical names and codes are then identified using the NUTS-based dictionary.

This prevents dates and locations from being added again to the semantic vocabulary. The remaining text is normalised and stored as the residual title.

For example, a title such as:

> Employment rate by sex and age in European regions, 2010–2020

would be reduced to a residual expression containing the main statistical concepts, while the date interval and geographical reference would be treated separately.

6.4 Residual title vocabulary

The residual title is added to the vocabulary associated with the table. It is preserved as a complete expression rather than being divided into isolated words.

This decision retains the semantic meaning of multi-word statistical concepts such as:

- employment rate by age;
- average household expenditure;

- population at risk of poverty;
- number of registered enterprises.

The title residual is included only when it contains meaningful content after normalisation.

6.5 Results

The system associated titles with [**NUMBER OF TABLES WITH TITLES**] tables. A total of [**NUMBER OF TABLES WITHOUT TITLES**] selected tables did not have an associated title.

During title processing, the system extracted [**NUMBER OF TEMPORAL EXPRESSIONS IN TITLES**] temporal expressions and [**NUMBER OF GEOGRAPHICAL EXPRESSIONS IN TITLES**] geographical references.

A total of [**NUMBER OF RESIDUAL TITLES**] residual title expressions were added to the table vocabularies.

Table 5 presents the main results.

Table 5: Results of table-title processing.

Result	**Value**
Tables with an associated title	7,604
Tables without an associated title	0
Temporal expressions removed from titles	1,284
Geographical expressions removed from titles	335
Residual titles added to the vocabulary	7,604

Title processing improves the coverage of the vocabulary because it introduces statistical concepts that may not appear directly in the table columns.

7 Construction of the Global Vocabulary

7.1 Aggregation process

After geographical filtering and title processing, every table contains a final local vocabulary $V(t)$.

The global vocabulary is constructed by combining the vocabularies of all processed tables:

$$V = \bigcup_t V(t) \tag{4}$$

Repeated terms are merged so that each normalised expression appears only once in the global vocabulary. However, the system preserves information about its occurrence across tables.

For each term, the final representation may include:

- normalised form;
- original textual form;
- number of tables in which it appears;

- total occurrence frequency;
- source type;
- associated table identifiers.

These features provide useful evidence for the later classification stage. For example, a term that appears repeatedly as a column header may be more likely to represent a dimension name, while a term extracted mainly from title residuals may be more likely to represent a measure.

7.2 Results

The final table-level vocabulary contained [**NUMBER OF FINAL TABLE-TERM PAIRS**] table-term pairs.

After merging duplicates across tables, the global vocabulary V contained [**NUMBER OF DISTINCT TERMS IN V**] distinct terms.

Table 6 summarises the size of the resulting vocabulary.

Table 6: Final global vocabulary results.

Result	**Value**
Final table-term pairs	745,418
Distinct terms in the global vocabulary	162,236

The resulting vocabulary is the input to the classification stage. At this point, dates and geographical entities have already been separated, but the remaining terms still contain different semantic roles. The next stage assigns each term to a vocabulary category.

8 Vocabulary Classification

8.1 Objective

The global vocabulary contains terms with different semantic functions. Some expressions represent quantities that can be measured, while others describe the dimensions used to organise the observations, the possible values of those dimensions or the units in which the data are expressed.

The objective of this stage is to partition the global vocabulary V into five categories:

- **Measures**: quantitative concepts or statistical indicators.
- **Dimension names**: labels that identify classification axes.
- **Dimension values**: categories that belong to a dimension.
- **Units**: measurement units, scales or reporting formats.
- **Other**: terms that cannot be assigned reliably to the previous categories.

Every term is assigned to exactly one category. Therefore, the final classification must satisfy:

$$|M| + |N| + |A| + |U| + |O| = |V| \tag{5}$$

where M, N, A, U and O represent measures, dimension names, dimension values, units and other terms.

8.2 Classification strategy

The classification method combines lexical rules, structural information and evidence collected during the previous stages.

Lexical rules identify characteristic expressions associated with each category. For example, measures frequently contain terms such as *rate*, *average*, *number*, *index*, *income* or *expenditure*. Units may contain percentages, currencies, physical quantities or expressions such as *per inhabitant* and *thousand persons*.

Dimension names are often short and general labels such as *sex*, *age*, *occupation*, *activity*, *unit* or *status*. Dimension values are more specific categories such as *female*, *employed persons*, *manufacturing* or *15 to 24 years*.

The origin of each term is also considered. Terms that repeatedly appear as column headers provide evidence for the dimension-name category. Terms found mainly as values inside descriptive columns are more likely to represent dimension values. Expressions obtained from table titles are more likely to describe measures or complete statistical concepts.

The classifier also uses term length, occurrence frequency, numerical patterns and contextual indicators. These signals are combined through ordered rules so that more specific evidence is applied before broader patterns.

8.3 Category definitions

8.3.1 Measures

Measures describe the phenomenon represented by the numerical observations of a table. They may refer to counts, rates, averages, percentages, indices or economic quantities.

Representative examples include:

- employment rate;
- number of enterprises;
- average annual income;
- population at risk of poverty;
- energy consumption.

Measures are frequently extracted from titles because titles usually describe the main subject of a table. They may also appear in descriptive headers or labels.

8.3.2 Dimension names

Dimension names identify the axes used to organise or filter the statistical observations.

Typical examples include:

- age;
- sex;
- occupation;
- economic activity;
- citizenship;
- unit.

These terms are usually short, generic and repeatedly used as column headers across different tables.

8.3.3 Dimension values

Dimension values are the possible categories associated with a dimension. This is normally the largest class because every dimension can contain many different values.

Examples include:

- males;
- 25 to 34 years;
- unemployed persons;
- agriculture;
- full-time employment;
- upper secondary education.

The classification of dimension values relies strongly on their appearance inside descriptive columns and on their relationship with known dimension names.

8.3.4 Units

Units specify how numerical observations are expressed. They include conventional measurement units, monetary units, ratios and reporting scales.

Examples include:

- percentage;
- euro;
- persons;
- tonnes;
- index, 2015 = 100;
- per thousand inhabitants.

Some expressions are ambiguous because words such as *persons* or *enterprises* can describe either a unit or a population category. Structural context and lexical patterns are therefore required.

8.3.5 Other terms

The category *Other* is used when the available evidence is insufficient or contradictory.

This category prevents weak classifications from being forced into one of the four main groups. Nevertheless, its use is intentionally limited because the objective is to classify as much of the vocabulary as possible.

8.4 Classification outputs

The system generates a complete file containing every term, its assigned category and the evidence used during classification.

It also generates one independent file for each category:

- `measures.csv`;
- `dimension_names.csv`;
- `dimension_values.csv`;
- `units.csv`;
- `other.csv`.

The separation of the categories allows each vocabulary subset to be analysed or reused independently.

8.5 Classification results

The global vocabulary contained [**NUMBER OF TERMS IN V**] distinct terms.

The classifier assigned:

- [**NUMBER OF MEASURES**] terms to the measure category;
- [**NUMBER OF DIMENSION NAMES**] terms to the dimension-name category;
- [**NUMBER OF DIMENSION VALUES**] terms to the dimension-value category;
- [**NUMBER OF UNITS**] terms to the unit category;
- [**NUMBER OF OTHER TERMS**] terms to the other category.

Table 7 summarises the final distribution.

Table 7: Distribution of the global vocabulary by category.

Category	**Number of terms**	**Percentage**
Measures	11,209	6.91%
Dimension names	15,204	9.37%
Dimension values	132,992	81.97%
Units	2,830	1.74%
Other	1	< 0.01%
Total	162,236	100%

Dimension values formed the largest category. This was expected because the tables contain many categorical values for age groups, activities, products, occupations, statuses and other classifications.

The measure category was smaller but particularly relevant because it represents the main statistical concepts and provides the input to the domain-assignment stage.

8.6 Manual evaluation

The quality of the classification was evaluated using a manually annotated sample of [**NUMBER OF EVALUATED TERMS**] terms.

Each sampled term was reviewed and assigned to its expected category. The manual labels were then compared with the automatic predictions.

The evaluation used accuracy, precision, recall and F1-score. Accuracy measures the proportion of correct predictions in the complete sample. Precision indicates how many terms assigned to a category actually belong to it. Recall indicates how many expected terms of a category were identified. The F1-score provides a balance between precision and recall.

The overall classification accuracy was [**CLASSIFICATION ACCURACY**]. A total of [**NUMBER OF CORRECT PREDICTIONS**] predictions were correct and [**NUMBER OF INCORRECT PREDICTIONS**] were incorrect.

Table 8 presents the results by category.

Table 8: Manual evaluation of vocabulary classification.

Category	**Precision**	**Recall**	**F1-score**	**Support**
Measures	[VALUE]	[VALUE]	[VALUE]	[VALUE]
Dimension names	[VALUE]	[VALUE]	[VALUE]	[VALUE]
Dimension values	[VALUE]	[VALUE]	[VALUE]	[VALUE]
Units	[VALUE]	[VALUE]	[VALUE]	[VALUE]
Other	[VALUE]	[VALUE]	[VALUE]	[VALUE]

8.7 Error analysis

The main classification errors were caused by terms whose role depends strongly on context.

Some population labels can be interpreted either as units or dimension values. For example, *persons* may describe the reporting unit of an observation, while *employed persons* normally represents a category or population group.

Short generic terms also create ambiguity. Expressions such as *total*, *value*, *status* or *number* do not always provide enough information when analysed independently.

Another source of error is the difference between a measure and a complete title expression. A long title may combine a measure with several dimensions, which makes it difficult to assign the complete residual phrase to a single category.

Despite these ambiguities, the evaluation indicates that the rule-based approach provides a consistent classification for most terms. The strongest results were obtained for categories with clear structural evidence, particularly dimension names. Units and dimension values were more difficult because their lexical forms overlap more frequently.

9 Grouping Measures into Statistical Domains

9.1 Objective

After vocabulary classification, the terms identified as measures still represent a heterogeneous collection of statistical concepts. Measures related to employment, health, education, energy or transport should not be treated as belonging to the same thematic group.

The objective of this stage is to assign every extracted measure to one statistical domain. This organisation makes the vocabulary easier to analyse and supports thematic search and comparison between measures.

The system uses seventeen domains:

- Agriculture, forestry and fisheries;
- Business, industry and trade;
- Crime, justice and safety;
- Demography and population;
- Economy and national accounts;

- Education and training;
- Energy;
- Environment and climate;
- Government and public finance;
- Health;
- Housing and construction;
- Income, poverty and living conditions;
- Labour market;
- Science, technology and digital society;
- Tourism;
- Transport and mobility;
- Other or multidisciplinary.

The final category is used for generic measures or expressions that do not contain enough evidence for a more specific thematic assignment.

9.2 Assignment method

The domain-assignment method combines lexical rules with TF-IDF-based similarity.

First, each measure is normalised and converted into a textual representation. Characteristic words and expressions are then compared with domain-specific vocabularies. For example, terms such as *employment*, *unemployment*, *occupation* and *working time* provide evidence for the labour-market domain. Similarly, terms such as *hospital*, *disease*, *mortality* and *health expenditure* indicate the health domain.

High-precision rules are applied before the similarity-based method. These rules are designed for expressions with a clear thematic interpretation. Applying them first reduces the risk that a measure with strong lexical evidence is assigned to a broader but less appropriate domain.

For the remaining measures, TF-IDF represents both the measure and the available domain vocabulary. The similarity between the measure and every domain is calculated, and the domain with the strongest score is selected.

The use of both methods provides a balance between precision and coverage. Rules perform well for clear expressions, while TF-IDF allows less obvious measures to be assigned using their lexical similarity with domain terminology.

9.3 Confidence levels

Every domain assignment is associated with a confidence level.

The system distinguishes three levels:

- **High confidence**: the assignment is supported by a specific lexical rule or strong similarity.
- **Medium confidence**: the measure contains useful evidence, but the distinction between possible domains is less clear.
- **Low confidence**: the measure is generic, multidisciplinary or only weakly related to the selected domain.

Confidence values do not represent a statistical probability. They are an interpretation of the amount and quality of lexical evidence available for the assignment.

Measures with insufficient domain evidence are assigned to *Other or multidisciplinary*. This prevents weak similarities from forcing a specialised classification.

9.4 Generated outputs

The domain-assignment stage produces three main files.

The file `measure_domains.csv` contains every measure, its assigned domain, its score and its confidence level.

The file `domain_summary.csv` contains the number of measures assigned to each domain and their average score.

The file `domain_examples.csv` contains representative measures for every domain and supports qualitative inspection of the results.

These files allow both a global analysis of the domain distribution and a detailed review of individual assignments.

9.5 Results

A total of [**NUMBER OF MEASURES**] measures were assigned to the seventeen available domains.

The system produced:

- [**NUMBER OF HIGH-CONFIDENCE ASSIGNMENTS**] high-confidence assignments;
- [**NUMBER OF MEDIUM-CONFIDENCE ASSIGNMENTS**] medium-confidence assignments;
- [**NUMBER OF LOW-CONFIDENCE ASSIGNMENTS**] low-confidence assignments.

Table 9 presents the final distribution of measures by domain.

Table 9: Distribution of measures by statistical domain.

Domain	**Measures**	**Average score**
Agriculture, forestry and fisheries	279	0.2105
Business, industry and trade	879	0.1953
Crime, justice and safety	135	0.1531
Demography and population	640	0.1495
Economy and national accounts	557	0.1674
Education and training	1,042	0.2354
Energy	180	0.2327
Environment and climate	204	0.1488
Government and public finance	217	0.1660
Health	336	0.1488
Housing and construction	158	0.1720
Income, poverty and living conditions	397	0.4643
Labour market	1,475	0.1692
Science, technology and digital society	1,266	0.1377
Tourism	138	0.3598
Transport and mobility	462	0.2545
Other or multidisciplinary	2,844	0.0139
Total	11,209	–

The largest category was *Other or multidisciplinary*, with 2,844 measures. This group mainly contains generic expressions and measures without enough lexical evidence for a specialised domain.

Among the specialised domains, the largest were *Labour market*, with 1,475 measures; *Science, technology and digital society*, with 1,266 measures; and *Education and training*, with 1,042 measures. Their size reflects the thematic distribution of the source collection rather than a difference in classification quality.

The system produced 5,851 high-confidence assignments, representing 52.20% of all measures. A further 2,240 assignments, or 19.98%, received medium confidence, while 3,118 assignments, or 27.82%, were classified with low confidence.

9.6 Manual evaluation

The domain assignments were evaluated using a manually annotated sample of [**NUMBER OF EVALUATED MEASURES**] measures.

Each sampled measure was reviewed and assigned to its expected domain. The manual annotation was then compared with the automatic prediction.

The domain classifier obtained an accuracy of [**DOMAIN ACCURACY**]. A total of [**NUMBER OF CORRECT DOMAIN PREDICTIONS**] assignments were correct and [**NUMBER OF INCORRECT DOMAIN PREDICTIONS**] were incorrect.

The macro F1-score was [**MACRO F1-SCORE**], while the weighted F1-score was [**WEIGHTED F1-SCORE**].

Table 10 summarises the evaluation results.

Table 10: Manual evaluation of measure-domain assignment.

Metric	Value
Evaluated measures	[VALUE]
Correct assignments	[VALUE]
Incorrect assignments	[VALUE]
Accuracy	[VALUE]
Macro F1-score	[VALUE]
Weighted F1-score	[VALUE]

9.7 Error analysis

The main domain-assignment errors were caused by measures that could be interpreted from more than one thematic perspective.

For example, employment-related measures may also appear in tables about education, poverty, industry or population. Similarly, public expenditure can belong to government finance, health, education or environmental policy depending on the context.

Another source of difficulty was the presence of generic measures such as *total value*, *number of observations* or *annual variation*. These expressions do not contain enough lexical information to identify a specific topic.

Long residual titles may also contain vocabulary from several domains. In such cases, the classifier selects the domain with the strongest evidence, although another interpretation may also be reasonable.

The difference between macro and weighted F1-scores must also be considered. The weighted score is influenced more strongly by larger domains, while the macro score gives the same importance to every represented domain. A lower macro score can therefore indicate that small or less frequent domains are more difficult to classify.

Overall, the evaluation shows that the hybrid method provides reliable assignments for most measures with clear domain terminology. The confidence levels and the *Other or multidisciplinary* category also make uncertain cases easier to identify for manual review.

10 Implementation and Processing Performance

10.1 Flexible table reading

The source collection contains files with different sizes, delimiters, encodings and compression formats. A single fixed reading configuration was therefore not sufficient for the complete dataset.

The implementation uses a flexible CSV-reading function that tests several common encodings and separators before selecting a valid configuration. The parser also detects whether a file is internally compressed with GZIP, independently of its filename extension.

This behaviour is important because several files use the `.csv` extension even though their content is compressed. Automatic detection avoids the need to rename or manually decompress every table before execution.

The reader is used consistently by the temporal-extraction and vocabulary-extraction stages. This reduces duplicated code and ensures that all modules handle heterogeneous files in the same way.

10.2 Processing of large tables

The largest tables cannot be loaded completely into memory. Vocabulary extraction therefore uses block-based processing.

After detecting the descriptive columns, the system reads only those columns in blocks of [**CHUNK SIZE**] rows. Numerical observation columns are not loaded because they are not required for vocabulary extraction.

This design provides two advantages. First, memory consumption remains stable even when the source file is very large. Second, the parser avoids processing most of the numerical content of the table.

Before the complete execution, the block-based procedure was tested on a table of approximately [**SIZE OF LARGE TEST TABLE**] GB. The test processed [**NUMBER OF BLOCKS**] blocks, identified [**NUMBER OF DESCRIPTIVE COLUMNS**] descriptive columns and extracted [**NUMBER OF DISTINCT TERMS**] distinct terms without exhausting the available memory.

10.3 Malformed and corrupted files

Some source files required special treatment.

One file, `aact_ali01.csv`, contained an incomplete internal GZIP stream. Reading the file produced an end-of-stream error, even after extracting it again from the original archive.

The integrity hash of the downloaded archive matched the expected value. This indicated that the local archive was complete and that the problem was associated with the internal file itself. Since the missing compressed data could not be reconstructed, the file was excluded through the

YAML configuration.

A second group of files contained malformed quotation marks. These files produced parsing errors because the quotation marks did not follow the standard CSV escaping rules. For these specific cases, quotation marks were treated as ordinary characters instead of field delimiters.

The affected files were:

- `prc_dap12.csv`;
- `prc_dap13.csv`;
- `prc_dap14.csv`;
- `prc_dap15.csv`.

This modification allowed their descriptive content to be processed without changing the behaviour of the parser for the rest of the dataset.

10.4 Configuration and reproducibility

The pipeline is controlled through a YAML configuration file. The main parameters include:

- dataset mode;
- input and output directories;
- accepted year range;
- block size;
- random seed;
- optional sample limit;
- list of excluded files;
- classification and clustering settings.

Two dataset modes are supported. The reduced mode is intended for development and reproducibility tests, while the full mode processes the complete collection.

All paths are relative to the project root. As a result, the code does not depend on the original development directory and can be executed on another computer after placing the required resources in the expected folders.

Intermediate and final results are stored in separate directories. This organisation makes it possible to inspect the output of every stage and simplifies debugging when an error occurs.

10.5 Execution performance

The complete pipeline was executed locally using Python 3.11. Internet access was not required during processing because all resources had already been downloaded.

The main factors affecting execution time were:

- CPU speed during GZIP decompression;
- disk read performance;
- number and size of the tables;
- parser overhead;
- selected block size.

The temporal-extraction stage was relatively fast because it only read table headers. Vocabulary extraction was the most expensive stage because it had to read the descriptive content of all selected tables.

The later stages worked mainly with the smaller intermediate CSV files generated by the pipeline and therefore required less processing time.

Table 11 summarises the final execution times.

Table 11: Execution time of the main processing stages.

Stage	**Execution time**
Temporal interval extraction	[VALUE]
Table vocabulary extraction	[VALUE]
Geographical term identification	[VALUE]
Title processing and vocabulary aggregation	[VALUE]
Vocabulary classification	[VALUE]
Measure-domain assignment	[VALUE]
Total execution time	[VALUE]

The final execution selected [**NUMBER OF SELECTED TABLES**] tables and completed the full pipeline in approximately [**TOTAL EXECUTION TIME**].

The results confirm that the block-based approach is suitable for the complete collection. The processing time is higher than for an in-memory solution on a small dataset, but the method avoids memory failures and supports tables that would otherwise be impractical to process on a conventional computer.

11 Overall Evaluation and Discussion

11.1 Evaluation procedure

The two stages that required the greatest semantic interpretation were evaluated manually: vocabulary classification and measure-domain assignment.

For vocabulary classification, a sample of terms was selected from the global vocabulary. Each term was manually assigned to one of the five expected categories and compared with the automatic prediction.

For domain assignment, a second sample was created using terms previously classified as measures. Each measure was manually associated with its expected statistical domain and compared with the domain selected by the system.

The evaluation samples were independent from the rules used to construct the classifiers. They were not used to adjust individual predictions after annotation.

The following metrics were calculated:

- **Accuracy**, representing the proportion of correct predictions;
- **Precision**, indicating how many predicted terms of a category were correct;
- **Recall**, indicating how many expected terms of a category were identified;
- **F1-score**, providing a balance between precision and recall.

For domain assignment, macro and weighted F1-scores were also calculated. The macro score gives the same importance to every represented domain, while the weighted score gives greater importance to domains with more examples.

11.2 Summary of evaluation results

Table 12 summarises the main evaluation results.

Table 12: Summary of the manual evaluation.

Evaluation task	**Sample size**	**Accuracy**
Vocabulary classification	[VALUE]	[VALUE]
Measure-domain assignment	[VALUE]	[VALUE]

The vocabulary classifier achieved its strongest results for dimension names. These terms frequently appear as column headers and usually have short and stable lexical forms.

Measures also obtained strong results because many of them contain characteristic words such as *rate*, *average*, *number*, *income* or *expenditure*.

Dimension values and units were more difficult to distinguish. Terms referring to persons, enterprises, products or activities can describe either a category or the unit used to report an observation.

The domain classifier performed well when the measure contained clear thematic vocabulary. Expressions related to employment, tourism, transport or education were generally assigned correctly.

Errors were more frequent for generic measures and concepts that could reasonably belong to several domains. For example, public expenditure may refer to government finance, education, health or environmental policy depending on the surrounding context.

11.3 Interpretation of confidence levels

The domain classifier assigns high, medium or low confidence to every prediction.

High-confidence assignments normally correspond to measures with direct lexical evidence or a specific matching rule. Medium-confidence predictions usually contain relevant words but could also be associated with another domain.

Low-confidence assignments represent the most uncertain cases. These include generic concepts, multidisciplinary measures and expressions whose title contains vocabulary from several domains.

The confidence information is useful because it allows uncertain predictions to be reviewed without manually inspecting the complete output. It also provides a more transparent result than assigning every measure to a domain without indicating the reliability of the decision.

11.4 Consistency checks

Several consistency checks were applied to the final outputs.

First, the sum of the five vocabulary categories must be equal to the total size of the global vocabulary:

$$|M| + |N| + |A| + |U| + |O| = |V| \tag{6}$$

Second, the sum of the measures assigned to the seventeen domains must be equal to the total number of extracted measures.

Third, the number of table-term pairs after geographical removal must be consistent with the number of geographical pairs removed from the initial vocabulary.

The final execution satisfied these checks:

- classified vocabulary total: [**VALUE**];
- global vocabulary total: [**VALUE**];
- measures assigned to domains: [**VALUE**];
- total extracted measures: [**VALUE**].

These checks do not guarantee that every individual term is semantically correct, but they confirm that the pipeline does not lose or duplicate terms between stages.

12 Limitations

12.1 Dependence on lexical rules

The classification system depends mainly on lexical and structural rules. This makes the method transparent and easy to inspect, but it also limits its ability to understand context in the same way as a manually curated system.

A rule may correctly classify most expressions containing a particular word while still failing for an unusual use of that word. The same statistical term may also have different roles depending on the table in which it appears.

12.2 Ambiguous vocabulary

Some terms cannot be assigned reliably without considering the complete table structure.

For example, *persons* may represent a unit, while *employed persons* may represent a population category. Similarly, *total* can be a dimension value, part of a measure or an aggregation label.

The system uses source information and lexical context to reduce this ambiguity, but some cases remain difficult.

12.3 Geographical coverage

Geographical recognition is based on the supplied NUTS workbook. This provides strong coverage for European territorial names and codes, but it may not include:

- historical geographical names;
- informal territorial expressions;
- non-European geographical entities;
- abbreviations not included in the source workbook.

As a result, some geographical terms may remain in the final vocabulary.

12.4 Title residuals

Residual titles preserve useful multi-word expressions, but they may also combine several semantic roles.

A title can contain a measure, one or more dimensions and a population group in the same phrase. Assigning the complete residual title to a single category can therefore introduce classification errors.

A more advanced system could divide titles into shorter semantic components before classification.

12.5 Domain overlap

The seventeen domains provide a practical organisation of the measures, but statistical topics are not completely independent.

Concepts such as employment, public expenditure, environmental taxes or health education can belong to several thematic areas. The current implementation assigns one main domain to every measure, even when a multi-label classification could be more appropriate.

12.6 Source-file quality

One internally compressed table could not be processed because its GZIP stream was incomplete. The archive integrity was verified, so the affected table was excluded from the final execution.

A small number of additional files contained malformed quotation marks. These files required a specific parsing configuration in which quotation marks were treated as ordinary characters.

These cases show that the quality and consistency of the source files influence the maximum coverage of the pipeline.

12.7 Evaluation size

The manually annotated samples provide evidence about the quality of the two classifiers, but they represent only a small part of the complete vocabulary.

A larger evaluation sample would produce more stable estimates and provide better coverage of rare categories and domains. However, manual annotation is time-consuming and requires understanding the context of the statistical terms.

Despite these limitations, the evaluation results indicate that the system provides a useful and consistent organisation of the vocabulary for most terms.

13 Conclusions

This project developed an end-to-end pipeline for extracting and organising statistical vocabulary from a large collection of Eurostat tables.

The pipeline separates temporal expressions, descriptive terms and geographical entities before constructing a global vocabulary. The resulting terms are classified into measures, dimension names, dimension values, units and other expressions. Measures are then grouped into seventeen statistical domains using a combination of lexical rules and TF-IDF similarity.

The final execution selected [**NUMBER OF SELECTED TABLES**] tables from the original collection. A total of [**NUMBER OF SUCCESSFULLY PROCESSED TABLES**] tables were processed successfully, while [**NUMBER OF EXCLUDED OR FAILED TABLES**] tables were excluded or could not be processed.

The global vocabulary contained [**NUMBER OF TERMS IN THE FINAL VOCABULARY**] distinct terms. These terms were divided into [**NUMBER OF MEASURES**] measures, [**NUMBER OF DIMENSION NAMES**] dimension names, [**NUMBER OF DIMENSION VALUES**] dimension values, [**NUMBER OF UNITS**] units and [**NUMBER OF OTHER TERMS**] other terms.

The vocabulary classifier achieved an accuracy of [**CLASSIFICATION ACCURACY**] on a manually annotated sample. The measure-domain classifier obtained an accuracy of [**DOMAIN ACCURACY**], with a macro F1-score of [**MACRO F1-SCORE**] and a weighted F1-score of [**WEIGHTED F1-SCORE**].

These results indicate that the proposed approach provides a useful automatic organisation of statistical terminology. The strongest classifications were obtained when the terms contained clear lexical or structural evidence. Ambiguous terms, generic expressions and multidisciplinary measures remained the main sources of error.

From a technical perspective, the use of block-based reading made it possible to process large and internally compressed tables without loading the complete dataset into memory. The modular structure also allowed each processing stage to be inspected and evaluated independently.

The project therefore meets its main objective of transforming heterogeneous statistical tables into a structured and reusable vocabulary. Although the result does not replace a manually curated ontology, it provides a practical starting point for statistical metadata management, semantic search and further vocabulary refinement.

13.1 Future work

Several improvements could be considered in future versions of the system.

First, title residuals could be divided into shorter semantic components. This would reduce the number of expressions that combine a measure, dimensions and population groups in the same term.

Second, the vocabulary classifier could use additional table context, including the relationship between column headers and their values. This could improve the distinction between units and dimension values.

Third, domain assignment could be extended from single-label to multi-label classification. This would be useful for measures related to several thematic areas.

The evaluation could also be extended with larger manually annotated samples and greater representation of rare categories and domains.

Finally, semantic relations such as broader, narrower and related terms could be generated and manually validated. This would transform the classified vocabulary into a more complete semantic resource.

14 Reproducibility

The source code is organised as a public and reproducible Python project.

The required software environment can be created using Python 3.11 and the dependencies listed in `requirements.txt`. The original datasets are not included in the repository because of their size, but the expected folder structure and file locations are described in the project README.

The pipeline is executed from the root directory using:

```
python main.py
```

Its behaviour is controlled through `config/config.yaml`. The configuration defines the selected dataset, paths, accepted year range, block size, exclusions and classification parameters.

A reduced dataset mode is available for testing the complete workflow without processing the full collection. The full mode reproduces the results presented in this report when the complete resources are placed in the required directories.

The random seed is fixed to **42** for procedures that require random sampling. Manually annotated evaluation files are stored separately and are not regenerated during normal execution.

The public repository is available at:

https://github.com/JuanMiguelPinos/statistic-vocabularies.git

The repository contains:

- the complete source code;
- the YAML configuration;
- installation and execution instructions;
- the dependency list;
- manually annotated evaluation samples;
- evaluation results;
- final output files compatible with repository size limits;
- this report.

The original Eurostat tables, compressed archives and large intermediate files are excluded from version control.

15 Use of AI Tools

AI tools were used during the development of the project as support for code review, debugging, documentation and the discussion of implementation alternatives.

Their main uses included:

- identifying possible causes of CSV and GZIP parsing errors;
- suggesting improvements for block-based processing;
- reviewing the organisation of the Python modules;
- assisting with the design of evaluation procedures;
- supporting the drafting and revision of the README and report.

The generated suggestions were not accepted automatically. All relevant code modifications were reviewed, executed and tested against the available datasets. Classification rules and evaluation labels were also inspected manually.

Representative prompts and a summary of the resources used during development are included in the repository documentation.

16 Final Project Status

The final project includes the following completed elements:

- extraction of temporal intervals;
- extraction of descriptive vocabulary;
- identification of geographical terms;
- processing of table titles;
- construction of the global vocabulary;
- vocabulary classification;
- grouping of measures by statistical domain;
- manual evaluation of the classification stages;
- scalable processing of the full dataset;
- public repository and reproducibility instructions.

The optional semantic-relation stage was [**IMPLEMENTED / NOT IMPLEMENTED**]. If implemented, its results are provided as candidate relations and should not be interpreted as a manually validated ontology.